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FOUNDATIONS OF MACHINE LEARNING

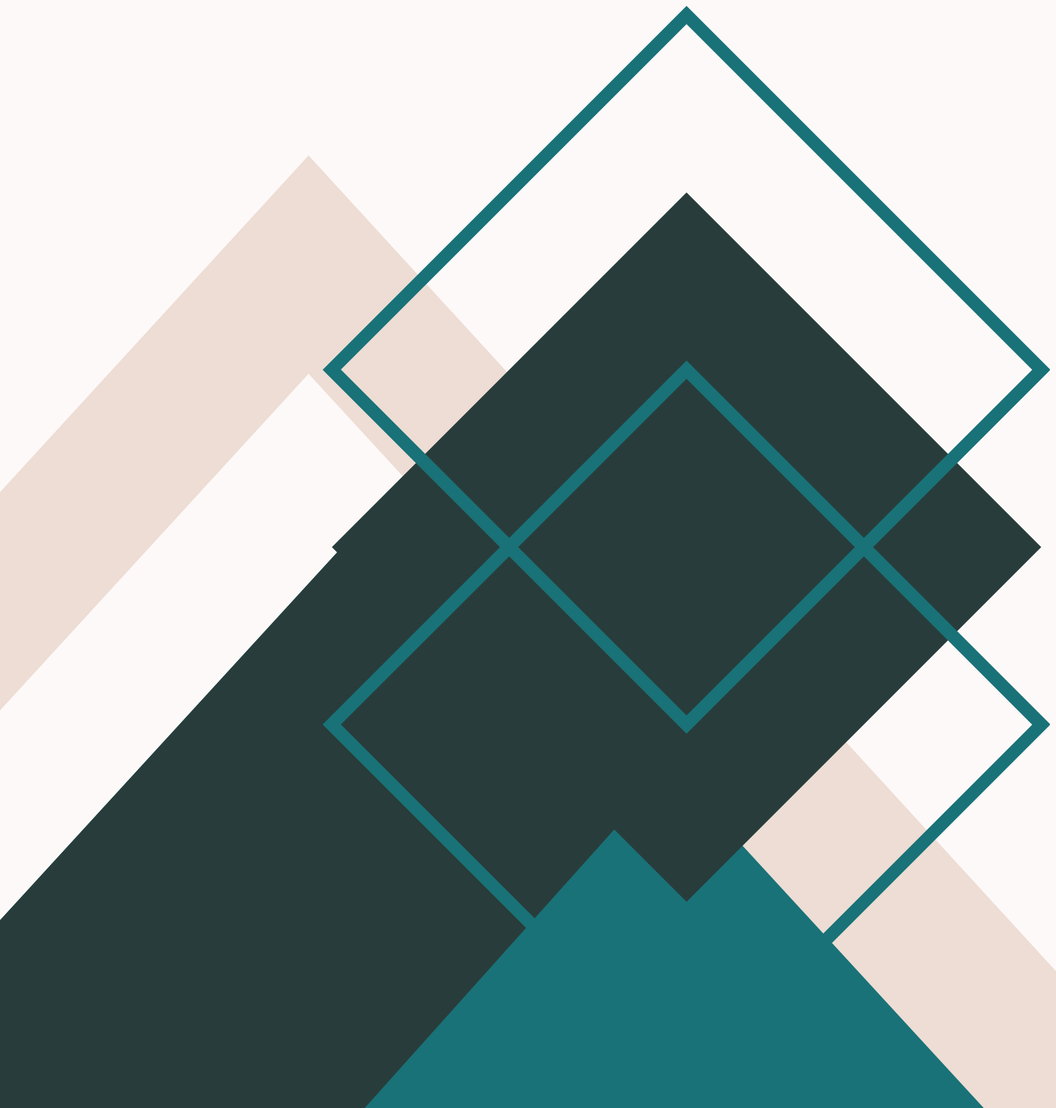
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INTRODUCTION

ShopSmart Retail Ltd. wants to begin integrating Artificial Intelligence into its retail operations.
Before developing advanced AI solutions, the company has hired AnalystLab Africa to build its first machine learning prototypes using historical retail data.

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OBJECTIVES

- Understand the ML lifecycle
- Prepare data for ML
- Build baseline ML models
- Evaluate model performance

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WHAT IS MACHINE LEARNING?

Machine Learning(ML) is a branch of Artificial Intelligence that enables computers to learn from data and make predictions or decisions without being explicitly programmed for every task. Similarly, ShopSmart Retail Ltd. can use historical retail data to predict the target variables:

1. Order_Total - total value of an order
2. Return_Flag - an estimation of whether an order is likely to be returned.

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MACHINE LEARNING LIFECYCLE



Problem Definition



Data Collection



Data Preprocessing & Cleaning



Feature Engineering



Model Selection & Training



Model Evaluation



Deployment & Integration



Monitoring & Maintenance

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DATA PREPARATION

Nine business datasets were loaded: customers, employees, inventory, orders, order items, products, returns, stores and suppliers. A data dictionary was also available. Orders and order items were merged using order and product identifiers, with a one-to-one validation for transaction rows. Customer, product and store attributes were joined into a 50,00-row, 37- column master dataset. Employee, inventory and supplier data were retained separately because a reliable direct transaction-level join was not established in the workflow.

Line_Total (A column in the master dataset) was calculated as Quantity x Unit_Price – Discount. For this assignment, Order_Total was defined as Line_Total + Shipping_Cost. This is a project assumption because no specific formula was supplied.

Return_Flag was set to 1 when an order ID appeared in the returns data and 0 otherwise. The returns table had 2,500 rows and 2,442 unique order IDs. The data-quality checks recorded no missing cells or exact duplicate rows in the merged master dataset.

Date fields were converted to datetime. The data was split into 80% training and 20% test sets, stratified by Return_Flag. Categorical features were one-hot encoded; numerical features were standerdised. The encoder and scaler were fitted on training data and applied to test data.

The selected features included Customer_Segment, Payment_Method, Sales_Channel, Category, Region, Quantity, Unit_Price, Discount and Shipping_Cost. The engineered regression model also used Quantity_Price = Quantity x Unit_Price.

The required CSV outputs are data/master_dataset.csv, data/train.csv and data/test.csv.

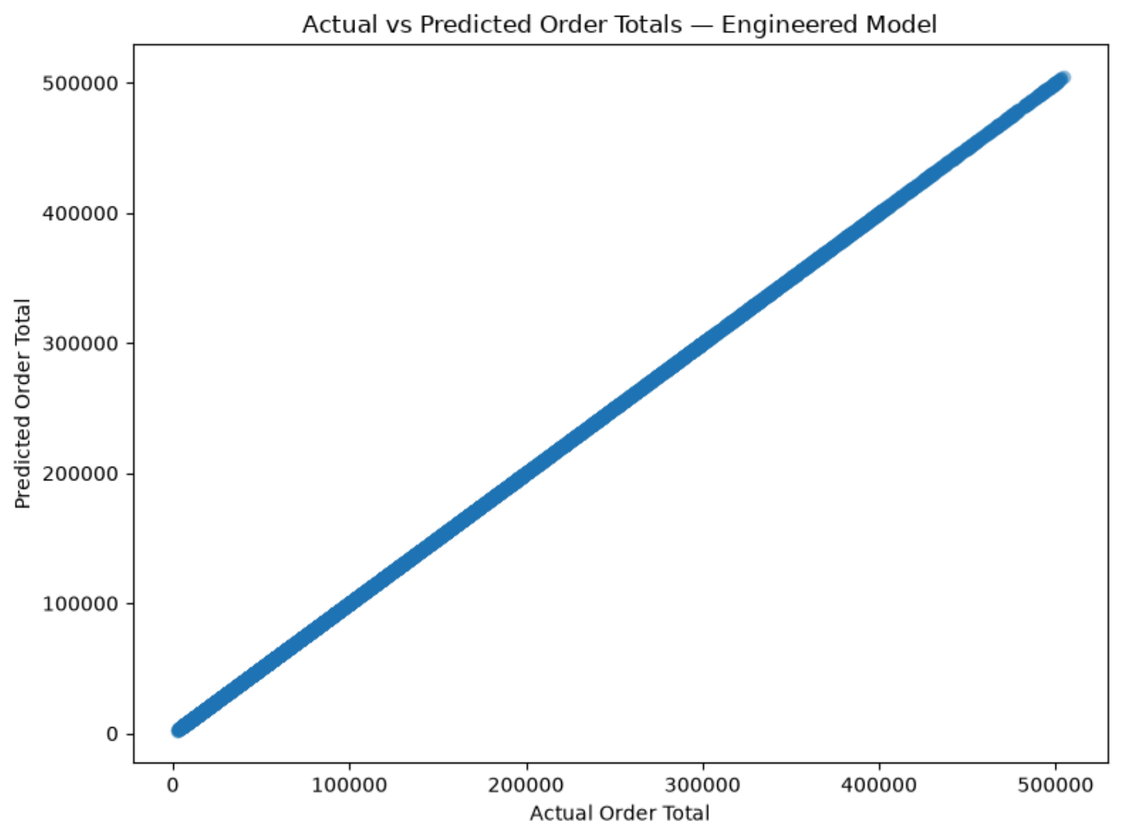
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MODEL DEVELOPMENT

Linear Regression

A baseline Linear Regression model was trained to predict Order_Total using encoded categorical variables and four numerical variables. An engineered version added Quantity_Price. The model was fitted on the transformed training matrix and evaluated on the held-out test matrix.

MODEL VISUALISATION & METRICS





Model	MAE	RMSE	R^2
Baseline Linear Regression	29,538.14	40,079.22	0.8875
Engineered Linear Regression	Approximately 0	Approximately 0	1.0000

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EVALUATION RESULTS: LINEAR REGRESSION

The Linear Regression model was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination R^2.
The initial model achieved an MAE of 29,538.14, an RMSE of 40,079.22, and an R^2 score of 0.8875. To improve its performance, a new feature, `Quantity_Price`, was engineered by multiplying `Quantity_order` by `Unit_Price_order`.
After adding this feature, the improved model achieved an MAE of approximately 0, an RMSE of approximately 0, and an R^2 score of 1.0000 on the test set. The actual-versus-predicted plot showed near-perfect alignment with the reference line, while the residual plot showed errors extremely close to zero.
The improvement occurred because the engineered feature represented the multiplication component of the formula used to calculate `Order_Total`. Therefore, the model successfully reproduced the defined target formula. However, this perfect performance should be interpreted in context: the target was constructed directly from the input features, so the result demonstrates formula reconstruction rather than necessarily demonstrating the ability to predict uncertain real-world order totals.

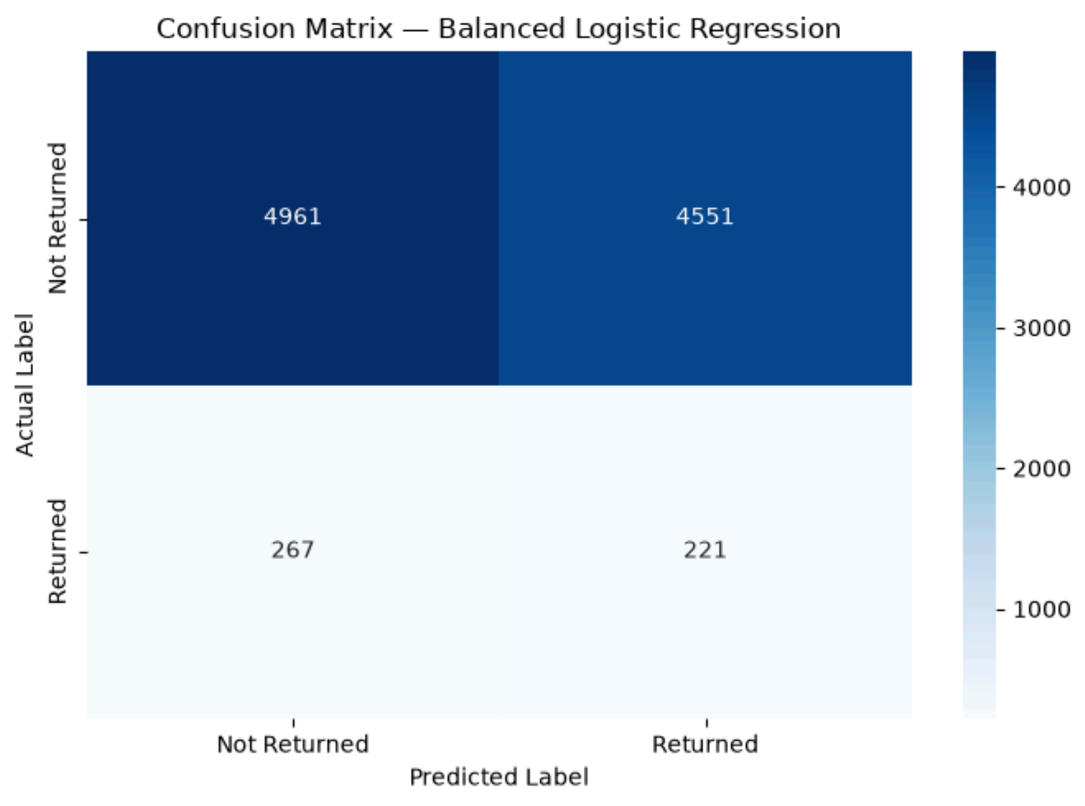
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CLASSIFICATION MODEL

Logistic Regression

A Logistic Regression model was trained to predict Return_Flag. Since returned orders were a minority class, a standard model was compared with a class-weighted model using class_weight='balanced'. The class-weighted model assigned greater importance to the minority class.
The models and preprocessing objects were saved with Joblib. Both saved models were loaded and generated 10,000 predictions each. The loaded predictions matched the original models' predictions on the same prepared test matrices, verifying successful saving and reloading.

MODEL VISUALISATION & METRICS



Metric	Baseline Logistic Regression	Class-weighted Logistic Regressio
Accuracy	0.9512	0.5182
Returned-class precision	0.0000	0.0463 (4.6%)
Returned-class recall	0.0000	0.4529 (45.3%)
Returned-class F1	0.0000	0.0844 (8.4%)

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EVALUATION RESULTS: LOGISTIC REGRESSION

The target variable was Return_Flag, where 0 represents an order that was not returned and 1 represents an order that was returned. A Logistic Regression model was trained using the prepared training dataset and evaluated on the test dataset. The initial model achieved an accuracy of **95.12%**. However, its precision, recall, and F1-score for returned orders were all **0.00**. This indicated that the model predicted every test order as not returned.

The dataset was imbalanced: approximately **95.1%** of orders were not returned, while only **4.9%** were returned. As a result, the model could achieve high accuracy by predicting the majority class without identifying any returned orders.

To address this issue, a second Logistic Regression model was trained using `class_weight="balanced"`. This option gives greater weight to the minority class during training. The balanced model achieved an accuracy of approximately 51.82%. Its precision for returned orders was approximately 4.63%, recall was approximately 45.29% and F1-score was approximately 8.44%. The model correctly identified 221 returned orders but incorrectly flagged 4,551 orders that were not returned. It also missed 267 actual returns.

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RECOMMENDATIONS

- **Redesign the order-value target**. For genuine forecasting, predict future spending or order value using only information available before the outcome is known; avoid features that directly recreate the target formula
- I**mprove return-risk modelling**. Evaluate classification thresholds and alternative approaches using precision, recall, F1 and precision-recall trade-offs. Select thresholds based on the costs of missed returns and unnecessary reviews.
- **Use the classifier only as decision support**. Given its low precision, test any review process in a controlled pilot rather than automatically acting on every flagged order.
- **Collect richer data**. Useful additions may include purchase history, order frequency, customer tenure, return reasons, product defects, delivery delays, promotions, reviews, stock availability and store-level costs or profit.
- **Validate over time**. A time-based holdout and ongoing monitoring can show whether performance generalizes to future periods.
- **Keep preprocessing consistent**. For new data, preserve feature order and use the same encoder, scaler and engineered-feature definitions as in training.
- **Report business measures carefully**. Distinguish sales value from profit association from causation and formula reconstruction from predictive performance.

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THANK YOU

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